

Daily File Namer

Code:

```
from airflow import DAG # type: ignore
from airflow.providers.standard.operators.python import PythonOperator # type: ignore
# from airflow.operators.bash import BashOperator
from airflow.providers.standard.operators.bash import BashOperator
from datetime import datetime

def name_the_file_func(**context):
    ds = context['ds']
    ds_nodash = context['ds_nodash']

    print(f"Naming file for date: {ds}")

    file_path = f"/reports/{ds_nodash}_daily_report.csv"
    print(file_path)

    return file_path

def explain_execution_date_func(**context):
    ds = context['ds']
    yesterday_ds = context['macros'].ds_add(ds, -1)

    print(f"1. This run's ds is: {ds}")
    print("2. ds is one day behind because Airflow only processes data AFTER the day is finished.")
    print(f"3. So this pipeline is processing data from: {yesterday_ds}")

with DAG(
    dag_id="daily_file_namer",
    schedule="0 8 * * *",
    start_date=datetime(2024, 1, 1), # every day at 8 AM
    catchup=False
) as dag:

    show_template_vars = BashOperator(
        task_id="show_template_vars",
        bash_command="""
        echo "Run date: {{ ds }}"
        echo "Date no dashes: {{ ds_nodash }}"
        echo "Yesterday: {{ macros.ds_add(ds, -1) }}"
        echo "Tomorrow: {{ macros.ds_add(ds, 1) }}"
        """
    )
```

```

    echo "DAG: {{ dag.dag_id }}"
    echo "Task: {{ task.task_id }}"
    echo "Run ID: {{ run_id }}"
    ""
)

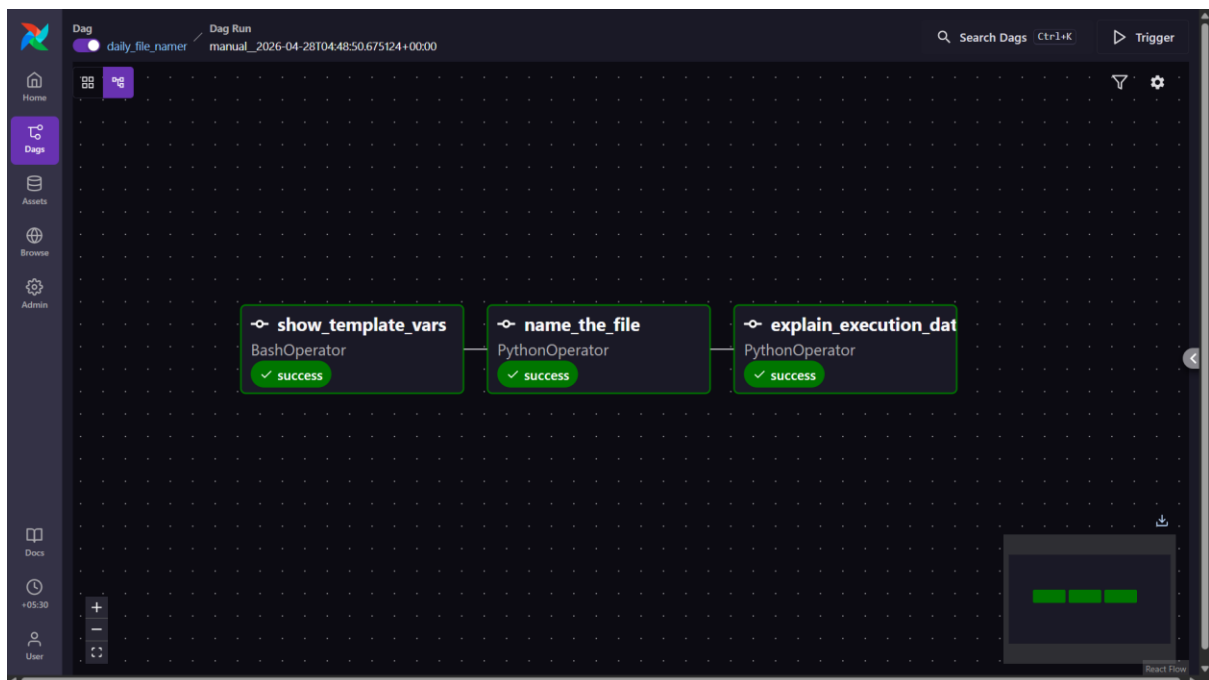
name_the_file = PythonOperator(
    task_id="name_the_file",
    python_callable=name_the_file_func
)

explain_execution_date = PythonOperator(
    task_id="explain_execution_date",
    python_callable=explain_execution_date_func
)

show_template_vars >> name_the_file >> explain_execution_date # type:
ignore

```

1. Graph View — all 3 tasks GREEN in order: show_template_vars → name_the_file → explain_execution_date



2. Task Log of show_template_vars — BashOperator output showing all 7 variables with real resolved values

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, Admin, Docs, and a clock showing 05:30. The main area shows a DAG named 'daily_file_namer' with a 'manual' trigger. The task 'show_template_vars' is highlighted with a green 'Success' status. The task details panel on the right shows the operator as 'BashOperator', start and end dates as '2026-04-28 10:18:51', duration as '00:00:00.114', and DAG version as 'v2'. The 'Logs' tab is active, showing a list of log messages. A yellow box highlights the output of the task, which includes the following log messages:

```
[2026-04-28 10:18:51] WARNING - Astro managed secrets backend is disabled
[2026-04-28 10:18:51] INFO - DAG bundles loaded: dags-folder
[2026-04-28 10:18:51] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/file_namer.py
[2026-04-28 10:18:51] INFO - TaskInstance Details: dag_id=daily_file_namer task_id=show_template_vars dagrun_id=manual_2026-04-28T04:48:50.675124+00:00
[2026-04-28 10:18:51] INFO - Temp dir root location: /tmp
[2026-04-28 10:18:51] INFO - Running command: ['/usr/bin/bash', '-c', '\n echo "Run date: 2026-04-28"']
[2026-04-28 10:18:51] INFO - Output:
[2026-04-28 10:18:51] INFO - Run date: 2026-04-28
[2026-04-28 10:18:51] INFO - Date no dashes: 20260428
[2026-04-28 10:18:51] INFO - Yesterday: 2026-04-27
[2026-04-28 10:18:51] INFO - Tomorrow: 2026-04-29
[2026-04-28 10:18:51] INFO - DAG: daily_file_namer
[2026-04-28 10:18:51] INFO - Task: show_template_vars
[2026-04-28 10:18:51] INFO - Run ID: manual_2026-04-28T04:48:50.675124+00:00
[2026-04-28 10:18:51] INFO - Command exited with return code 0
[2026-04-28 10:18:51] INFO - Pushing xcom ti=RuntimeTaskInstance(id=UUID('019dd26b-25fe-7cd2-9100-3c907f4d92
```

3. Task Log of explain_execution_date — your 3 printed lines using the actual ds and yesterday_ds values

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, Admin, Docs, and a clock showing 05:30. The main area shows a DAG named 'daily_file_namer' with a 'manual' trigger. The task 'explain_execution_date' is highlighted with a green 'Success' status. The task details panel on the right shows the operator as 'PythonOperator', start and end dates as '2026-04-28 10:18:53', duration as '00:00:00.103', and DAG version as 'v2'. The 'Logs' tab is active, showing a list of log messages. A yellow box highlights the output of the task, which includes the following log messages:

```
[2026-04-28 10:18:53] WARNING - Astro managed secrets backend is disabled
[2026-04-28 10:18:53] INFO - DAG bundles loaded: dags-folder
[2026-04-28 10:18:53] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/file_namer.py
[2026-04-28 10:18:53] INFO - TaskInstance Details: dag_id=daily_file_namer task_id=explain_execution_date dagrun_id=manual_2026-04-28T04:48:50.675124+00:00
[2026-04-28 10:18:53] INFO - 1. This run's ds is: 2026-04-28
[2026-04-28 10:18:53] INFO - 2. ds is one day behind because Airflow only processes data AFTER the day is finished.
[2026-04-28 10:18:53] INFO - 3. So this pipeline is processing data from: 2026-04-27
[2026-04-28 10:18:53] INFO - Done. Returned value was: None
```